Publication Decision from *Quantum*

"Floquet Topological Frequency-Converting Amplifier"

Decision made on August 19, 2026

Editorial board's determination

Revise and resubmit

Comments from the editor

Dear Adrian Parra Rodriguez,

We have received the referee reports for your paper “Floquet Topological Frequency-Converting Amplifier”. You may find them enclosed.

The referees are generally positive about the technical quality and potential interest of the work, with one referee recommending acceptance. Before the manuscript can be considered further, however, we ask you to address the main concern raised by the other referee regarding the conceptual novelty and central physical message of the work. In particular, the revised manuscript should make substantially clearer what distinguishes the proposed scheme from existing results on non-Hermitian topological amplification, Floquet synthetic dimensions, and singular-value topology—whether the key advance lies in the minimal single-mode realization, the simultaneous modulation of frequency and dissipation, or in a genuinely distinct amplification mechanism.

Please also address the referee’s specific technical comments and clarify the experimentally observable signatures of the proposed mechanism, including the possible role of the non-Hermitian skin effect.

Please upload the **revised version** and a **response letter file** to the points raised by the referees through Scholastica. It is in your own interest to ensure that the Referees and Editor can easily retrace your revisions (e.g., by attaching a detailed list of changes or a version of the manuscript with the changes highlighted). There is no need to update the arXiv version at this stage. If you have **confidential information** for the editor, please upload it as a **separate file**.

Best regards,
Angelo

Angelo Carollo
Editor at Quantum

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Reviewer 1

Open response questions

Summary: what are the main questions posed by the manuscript and how does it answer them?

Please see attached file

What is your assessment of the paper? If you recommend acceptance, make a case that this work does indeed make a significant contribution to scholarship.

n/a

To what extent have you checked the technical correctness of the submission? This includes for example examining mathematical proofs or source code.

n/a

Comment on the presentation of the paper. Is it well written? Are the main results clearly laid out? Does the manuscript clearly describe assumptions and limitations? Is the literature review adequate?

n/a

If the submission includes software, numerical or physical experiments, does it provide sufficient details such that they could be reproduced by readers? This includes for example source code, documentation, experimental data, experimental setup specifications, etc.

n/a

Suggested changes, corrections, and general comments.

n/a

Reviewer attachments

347819-anonymized-attachment-for-review-595484.pdf (https://app.scholasticahq.com/api/v1/attachments/347819/download)

Reviewer 2

Open response questions

Summary: what are the main questions posed by the manuscript and how does it answer them?

In this manuscript the authors consider a single driven–dissipative bosonic mode whose frequency and decay rate are periodically modulated. They show that in synthetic/frequency space the system can be mapped to a non-Hermitian tight-binding chain with nonreciprocal nearest-neighbour hopping and a synthetic electric field. Through a doubled-space mapping, they extract from the (Floquet) Green's function a local winding number that predicts the amplification and up/down-conversion between harmonics.

What is your assessment of the paper? If you recommend acceptance, make a case that this work does indeed make a significant contribution to scholarship.

I found the central ideal of the manuscript, i.e., modulating both coherent and dissipative terms to generate topological amplification in an otherwise zero-dimensional platform, original. In particular, I found the mapping to

the Jackiw-Rebbi model extremely elegant. The quality of this research contribution is high and manuscript the is extremely well written.

To what extent have you checked the technical correctness of the submission? This includes for example examining mathematical proofs or source code.

I inspected results, figures and equations but I did not reproduced any calculations myslef.

Comment on the presentation of the paper. Is it well written? Are the main results clearly laid out? Does the manuscript clearly describe assumptions and limitations? Is the literature review adequate?

see above

If the submission includes software, numerical or physical experiments, does it provide sufficient details such that they could be reproduced by readers? This includes for example source code, documentation, experimental data, experimental setup specifications, etc.

n/a

Suggested changes, corrections, and general comments.

Before recommending it for publication, I have the following questions:

- There is no characterisation of the added noise. This is my main criticism. I understand that a full analysis can be a project of its own and can be deferred to later works, but providing an estimate or a scaling argument would be extremely useful.

- Against which imperfections does a nonzero winding number provide protection in this specific single-mode realization?

- Related question: the construction assumes a common modulation frequency of the oscillator and the loss in Eqs. (3) (4). Are the results robust with respect to an imbalance between the two?

- Contrary to Hatano-Nelson model, in the topological phase localisation is in the bulk at +/- n_0, as made clear by the Jackiw-Rebbi mapping. I guess this is due to the synthetic electric field. Can the authors comment on that?

- Contrary to a spatially extended lattice, in the proposed implementation assuming periodic boundary conditions has not an obvious operational meaning. Can the authors comment on that?

- I find that Figure 2 packs a lot of information, perhaps too much, which is not easy to extract. Panels (e), (f) are quite hard to read. Why does the inset of panel (d) predict a non-trivial winding number between $-19 < n < 19$?

- I think it would be useful to the reader to add to the literature the (not so many) works that considered time-modulating dissipative rate, in the spirit of Floquet Lindbladian, see e.g. A. Schnell et al, Phys. Rev. B 101, 100301(R) (2020), A. Keliri, M. Schirò, arXiv:2606.09727.